

Anatoly Zavyalov

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Research Interests

Differential privacy, property testing, graph algorithms, sublinear algorithms

Education

Boston University

Ph.D. Student in Computer Science — Advised by Sofya Raskhodnikova

Sep 2024 - Jun 2029 (expected)

University of Toronto

H.B.Sc. in Applied Mathematics — 3.95 cGPA

Sep 2020 - Jun 2024

Selected Publications

1. S. Raskhodnikova, A. D. Smith, C. Wagaman, **A. Zavyalov**, “Local Node Differential Privacy”. arXiv preprint [arXiv:2602.15802](https://arxiv.org/abs/2602.15802) (2026).
2. J. Shallit, **A. Zavyalov**. “Transduction of Automatic Sequences and Applications”. CIIA 2023. Lecture Notes in Computer Science, vol 14151.
3. A. D. Hincks, **A. Zavyalov**, D. Bansal. “A graph database solution for tracking the deployment and layout of a large radio interferometer”. *SPIE Astronomical Telescopes + Instrumentation 2022*. Proc. SPIE 12189.

Honors and Awards

Chair’s Graduate Fellowship - \$10,000 USD

Awarded on admission to Boston University.

Feb 2024

NSERC Undergraduate Student Research Award - \$7,500 CAD

Awarded by U of T Department of Computer Science.

Mar 2023

SURP Fellowship & U of T Excellence Award - \$17,095 CAD

Combined funding for a research fellowship at the Dunlap Institute of Astronomy.

Apr-May 2021

NSERC USRA (\$7,500 CAD) & Fields USRP — Declined

Received and declined offers for two separate summer research programs.

Mar 2022

University of Toronto Scholar - \$3,000 CAD

Awarded by the University of Toronto for high academic achievement (\$1,500 × 2).

Aug 2021, Aug 2022

Dean’s List Scholar

Awarded for a cumulative GPA of 3.50 or higher.

2021, 2022, 2023

Trinity College Academic Achievement Scholarships - \$1,500 CAD

Awarded by Trinity College for high academic achievement (\$500 × 3).

Nov 2021 - Nov 2023

Talks

1. “Local Node Differential Privacy”: University of Oxford (March 2026), Columbia University (April 2026).
2. “Transduction of Automatic Sequences and Applications”, CIIA 2023. (30 min) ([Slides](#))
3. “Automatic Sequences”, Canadian Undergraduate Math Conference, 2023 (30 min) ([Slides](#))

Research Experience

Boston University | PhD Student

Sep 2024 - Present

Department of Computer Science

- Working in the areas of graph differential privacy and graph property testing, supported by a **Chair's Graduate Fellowship**.

University of Waterloo | Research Assistant

May 2022 – Jul 2022, May 2023 – Jul 2023

David R. Cheriton School of Computer Science

- Researched and implemented algorithms into **Walnut**, a theorem proving software for automatic sequences written in **Java**, working with Professor Jeffrey Shallit.
- Research culminated in a publication and presentation at *CIAA 2023*.

University of Toronto | Research Assistant

May 2023 – Aug 2023

Department of Computer Science

- Researched algebraic methods for concurrent program verification and race condition detection, supported by an **NSERC Undergraduate Student Research Award (NSERC USRA)**.

University of Toronto | Research Fellow

May 2021 – Apr 2022

David A. Dunlap Department of Astronomy and Astrophysics

- Developed **Padloper**, a full-stack graph database solution for tracking deployment and layout of a large radio interferometer, using **JanusGraph**, **Flask** and **React**, working with Professor Adam Hincks.
- Padloper will be used for the **Hydrogen Intensity and Real-time Analysis eXperiment (HIRAX)**, the **Simons Observatory**, and **Canadian Hydrogen Intensity Mapping Experiment (CHIME)**.
- Research culminated in proceedings at *SPIE Astronomical Telescopes + Instrumentation 2022*.

Teaching Experience

Boston University | Teaching Fellow

- Teaching Fellow for **CSC237: Probability in Computing**

Sep 2025 - Dec 2025

University of Toronto | Teaching Assistant

- TA for **CSC363: Computational Complexity and Computability**

Jan 2024 - Apr 2024

- Lead TA for **CSC373: Algorithm Design and Analysis**

Sep 2023 - Dec 2023

- TA for **CSC240: Enriched Introduction to the Theory of Computation**

Feb 2022 - May 2022

STEM Outreach

SigmaCamp | Academic Administrator & Counselor

2022 - Present

- Directing *Problem of the Month*, a free international STEM competition for teenagers, which received over 2,000 solution submissions from 130+ students in 2024-2025.
- Assembled and managed a team of 12 subject leads and dozens of graders. Also managed the Qualification Quiz entrance exam, which received 650+ solutions submissions in 2025.

Skills

- Human Languages:** English (C2), Russian (C2), German (B1), Portuguese (A2)
- Programming Languages:** Python, JavaScript, C++, Java, PostgreSQL, $\text{\LaTeX}$
- Frameworks/Libraries:** ReactJS, Flask, NumPy, PixiJS